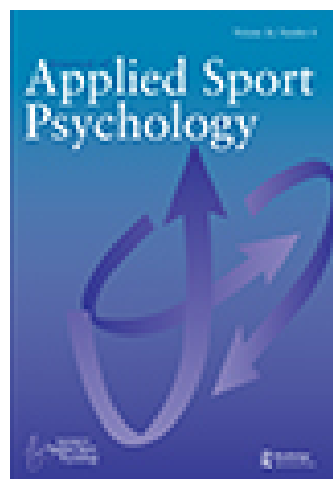


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Weight Training and Body Satisfaction of Body-Image-Disturbed College Women

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Body dissatisfaction and body image disturbances are prevalent in college age females and can contribute to reduced health and wellness. Therefore, it is important to understand factors that may be associated with these body image-related concerns. This study compared the body satisfaction of body-image-disturbed college women involved in weight training to those not involved. Body satisfaction of body-image-disturbed undergraduate females ($M_{\text{age}} = 21.5$, $SD = 4.32$) was assessed prior to and following a thirteen-week period of regular weight training involvement or non-involvement. Results of an ANCOVA, $F(2, 27) = 5.67$, $p < .05$, revealed that the mean satisfaction scores of body-image-disturbed weight trainers increased from 3.19 to 3.40 while the control group maintained their scores at 3.10. Approximately 41% of weight training participants displayed body image improvements such that they were no longer classified as body-image-disturbed. These results provide evidence to suggest that weight training may be an effective treatment for body image disturbance.

As health care costs escalate and life expectancy increases, there is increased attention on the health and wellness of all individuals. One area of focus that may be underestimated in its importance to health and well-being is body image. Body image refers to individuals' personal view of their own body (Rodin, 1992). Because body image represents individuals' perception of their body this physical self-perception may or may not reflect reality. Body image is of particular concern in a culture that tends to focus on body aesthetics as there are potent messages regarding society's tolerance of specific physical characteristics; there is considerable pressure for women to be thin and beautiful, and men to be muscular (Rodin, 1992; Stormer & Thompson, 1996; Thompson & Heinberg, 1993).

With the pressures of society, it is no surprise that, as noted by Zerbe (1993), it is common for people to be dissatisfied with their own personal body image. Though both men

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and women experience dissatisfaction with their bodies, Dolan, Birtchnell, and Lacey (1987) found the incidence of body dissatisfaction to be greater in women than men. Moreover, according to Hutchinson (1993), approximately 75% of American women wish their bodies looked different.

Body dissatisfaction can result from body image distortion, a misperception of one's appearance (see Dolan et al., 1987). It is prevalent in the United States, and is typically manifested in the overestimation of body size (Rodin, 1992). In fact, 95% of American women overestimate their body size by one fourth (Pipher, 1994). In addition, research indicates that approximately half of the women from teenagers to thirty-five years presume they weigh too much, and are striving to lose weight (Davis & Cowles, 1991; Dolan et al., 1987; Gardner, 1996; McAuley & Burman, 1993).

Although body image distortions are commonly associated with women who are anorexic or obese, women who fall within normal weight can also be victims of body image distortions, often perceiving themselves as larger than they are (Dolan et al., 1987). For example, Davis and colleagues (Davis, Fox, Cowles, Hastings, & Schwass, 1990) found that 77% of generally slender and active women wanted to lose weight, and most were actually dieting.

Body image distortion that results in negative body image and body dissatisfaction can have a debilitating effect on health and wellness. For example, females who have a negative body image and are dissatisfied with their bodies tend to have lower self-esteem than body-image-satisfied females (Berry & Howe, 2000; Zerbe, 1993). Of even greater concern, negative body image and dissatisfaction can lead to body image disturbance, an extreme dissatisfaction with one's body or one's preoccupation with an imagined imperfection in physical appearance (Cooper, Taylor, Cooper, & Fairburn, 1987).

Though a concern for women with or without disordered eating, body image disturbance plays a central role in eating disorders. Specifically, body image disturbance is thought to be a risk factor for the emergence of eating disorders and is considered one of the most important predictors of eating severity and weight aberrations among those with eating disorders (Horne, VanVactor, & Emerson, 1991). This seems to be associated with the abnormal eating behaviors and attitudes resulting from over concern about one's body (Horne et al., 1991). The influence of the emergence and severity of eating disorders, body image disturbance is a predictor of treatment response and relapse of eating disorders (Rosen, Cado, Silberg, Srebnik, & Wendt, 1990).

In addition to the implications with eating disorders, body image disturbance can be detrimental to health and well-being in other ways (Stormer & Thompson, 1996). It can disrupt normal functioning in relationships, and is associated with depression, social introversion, and poor self-esteem (Zerbe, 1993). Rosen and colleagues (1990) contend that body image disturbance qualifies as a psychological health matter that requires professional attention.

To date, the treatment of body image disturbance primarily involves cognitive therapy, which includes the use of automatic and rational thought processes, counterarguments, and rational beliefs strategies. Studies have shown that cognitive therapy treatment is effective in improving body image disturbance in college-aged, non-eating-disordered women (Hilyer & Mitchell, 1979; Rosen et al., 1990). This type of therapy can also lead to decreases in the distortion of body size, desire to be thinner, disruptive preoccupation with weight and shape, and dissatisfaction with appearance (Rosen et al., 1990).

In addition to cognitive therapy for body image disturbance, exercise may also be an especially appealing method for reducing negative body-related self-perceptions held by body-image-disturbed individuals. Two contemporary theories of self-esteem (Fox & Corbin, 1989; Sonstroem, 1997) suggest that physical activity can impact individuals' body perceptions.

Specifically, physical activity is thought to positively impact individuals' physical self-perceptions (e.g., strength and conditioning), leading to enhanced feelings of physical self-worth, which in turn results in enhanced global self-esteem. This contention is supported via numerous studies demonstrating that exercise, including aerobic and resistance training, improves the self-perceptions and body image of healthy males and females, independent of their body image prior to exercise (Bartlewski, Van-Raalte, & Brewer, 1996; Caruso & Gill, 1992; Eickhoff, Thorland, & Ansorge, 1983; Franzoi, 1986; Hayes, Crocker, & Kowalski, 1999; James, 1982; Tucker, 1982a, 1982b; Tucker & Maxwell, 1991).

The majority of body image research incorporating exercise has examined the effects of aerobic exercise (e.g., Bartlewski et al., 1996; Davis & Cowles, 1991; Eickhoff et al., 1983) and has shown it to be effective in enhancing body esteem and lowering social physique anxiety. A few studies have compared the relationship of aerobic physical activity and weight training on body satisfaction. The limited research that has been conducted in this area has demonstrated that weight training is associated with greater positive changes in body satisfaction (Caruso & Gill, 1992; Tucker & Mortell, 1993) and self-esteem (Trujillo, 1983) than aerobic exercise. For example, Tucker and Mortell examined the effects of a weight training vs. walking program on inactive, normal-weight women. Participants completed the Body-Cathexis Scale assessing their body satisfaction and were assigned to either a walking or a resistance-training group, and each group exercised accordingly for twelve weeks. Results revealed that the strength trainers demonstrated significantly greater improvements in body satisfaction than the walkers. Moreover, those in the resistance-training group with the poorest pre-training body satisfaction experienced the greatest gains in body satisfaction. The results of this study provide evidence to suggest that weight training can benefit individuals who are most dissatisfied with their bodies.

There are several possible mechanisms explaining improved body satisfaction and ultimately body image disturbances resulting from weight training. For example, the training environment provides individuals with positive feedback as the training load is progressively increased and recorded by the lifter to match the body's improving strength (Tucker, 1982a). Strength training also tends to produce more explicit changes in the body such as enhanced muscular strength, decreased total skinfold thickness, and increased lean body mass changes that may contribute to improvements in body image (Caruso & Gill, 1992; Tucker & Mortell, 1991). Additionally, changes in body physique occur relatively quickly in comparison to other forms of training (Tucker, 1982a, 1982b, 1987).

Not all body image research has supported the beneficial effects of exercise. For example, Sonstroem, Harlow, and Josephs (1994) found that women who exercised more often scored lower on the physical self-worth scale than women who did not participate in regular physical activity. Much of the research that has demonstrated the negative effects of exercise on self-perceptions has shown that *over-emphasis* on exercise can aggravate self-consciousness (Brumberg, 1988), particularly among high level athletes and high intensity exercisers. Brownell, Rodin, and Wilmore (1987) discovered that dieting and weight concerns were greatest among athletes at the highest levels of performance, especially in athletes whose bodies are an integral part of their performance evaluation such as ice skaters, gymnasts, and ballet dancers. For many, the concern can turn into obsession. Thirty-eight percent of female athletes surveyed stated they had eating disorder tendencies such as thoughts of purging and feelings of lacking control when eating (Brownell et al., 1987). In addition, some researchers suggest that it is the serious exercisers who are prone to body image disturbances and are most likely to show both physical and psychological symptoms of eating disorders (Hamilton, Brooks-Gunn, & Warren, 1985). Women who exercised more than six hours a week have been found to be more discontent with their bodies than those who exercised less than six hours a

week (Leveillce, Sonstroem, & Riebe, 1994). These women also displayed significantly higher scores on the Drive for Thinness, Bulimia and Body Dissatisfaction scales of the Eating Disorder Inventory. Collectively, this research reveals evidence to suggest a link between physical activity and body dissatisfaction among high intensity athletes and exercisers. Such a relationship between more moderate intensity anaerobic physical activity such as weight training and body dissatisfaction among body-image-disturbed women is less clear.

In summary, previous research exploring exercise and body image has focused primarily on aerobic exercise. More research is needed to examine the relationship between weight training and body image, especially among women with known body image disturbances. This type of investigation may prove beneficial in the development of guidelines for promoting more favorable body image among college women with body image disturbances, which in turn may result in enhanced health and well-being. Thus, the purpose of this study is to examine the effects of weight training on body satisfaction among college-age women with body image disturbances. It is hypothesized that body-image-disturbed women who participate in a weight training program will experience greater improvements in body satisfaction than those who do not participate in the weight training program. Although weight training was expected to relate to improvements in body satisfaction, potential changes in the classification of body-disturbance are unknown. Thus, the classification of individuals as body-image-disturbed following the study was exploratory and no hypotheses were forwarded.

METHODS

Participants

A total of 96 undergraduate females recruited from general education and weight training classes and ranging in age from 18 to 41 years ($M = 21.5$, $SD = 4.32$) agreed to participate in this study. Of these, 34 (35%) fit the criteria for body image disturbance (using the Body Shape Questionnaire) and 32 (94% of the eligible participants) agreed to participate in the experimental phase of the study. Fifteen of the participants were enrolled in general education courses and served as the control group, while 17 were enrolled in weight training classes and served as the experimental group. Two participants in the weight-training group withdrew from the study due to injuries. Therefore, the final sample consisted of 30 ($n = 15$ experimental group, $n = 15$ control group) body-image-disturbed females. Students in the weight training classes represented body-image-disturbed females who exhibited a readiness for exercise. Some of these participants were in the same weight training class, which leaves open the possibility of alternative hypotheses and limits the generalizability of the results.

The weight trainers ranged in age from 18–41 years with an average age of 22.13 years ($SD = 5.51$). Forty percent of the weight trainers were Caucasian, 20% African American, 13.3% Eastern Asian, 6.7% Mexican, 6.7% Filipino, and 13.3% did not report their ethnicity. Prior to their participation in the study, weight trainers exercised an average of 3.15 ($SD = 1.45$) days a week, but did not engage in weight training. The control group ranged in age from 18–23 years with an average age of 20.06 years ($SD = 1.33$). Approximately 86% were Caucasian and 13.3% did not report their ethnicity. They exercised an average of 2.23 ($SD = 2.02$) days per week and were not engaged in weight training.

Measures

All participants completed a series of questionnaires including the Body Shape Questionnaire, the Body-Cathexis Scale, and a background information questionnaire. The background information was used for demographic descriptors.

Body-Cathexis Scale

The Body-Cathexis Scale (BCS; Secord & Jourard, 1953) is a 46-item scale designed to assess body satisfaction. Participants responded on a 5-point Likert-type scale indicating their satisfaction with their own body, with 1 representing a person's "strong feelings and wish change could somehow be made" to 5 representing that a person "considers myself fortunate" regarding a specific body part or function. Mean scores for each individual were calculated by summing the ratings for all items and dividing the sum by the total number of answered items. Secord and Jourard (1953) administered the BCS to 56 college females and reported split-half reliabilities of 0.83 for females to show internal consistency. In addition, Tucker (1983) reported a test-retest reliability of 0.87 when using the BCS. The BCS also correlates with self-cathexis scores with reliabilities of 0.66 for females, indicating that individuals have a tendency to cathect their bodies to the same degree and direction that they cathect themselves. The Maslow Test of Psychological Security-Insecurity was also found to significantly correlate ($r = .37$) with the BCS (Secord & Jourard, 1957; Tucker, 1983, 1987; Tucker & Maxwell, 1991; Tucker & Mortell, 1993).

Body Shape Questionnaire

The Body Shape Questionnaire (BSQ; Cooper et al., 1987) is a 34-item, self-report questionnaire designed to measure body image disturbance, and includes questions regarding body dissatisfaction, fear of fatness, distressing preoccupation with appearance, feelings of low self-worth because of appearance, and the desire to lose weight. In completing the questionnaire, participants indicated how often they experienced negative body thoughts on a 5-point Likert-type scale, with 1 indicating *never* and 5 indicating *always*. The BSQ scores were calculated by summing the numerical equivalent of the responses. This questionnaire was utilized in this study to distinguish females with body image disturbances from body image non-disturbed females. The value used to distinguish body-image-disturbed from non-body-image-disturbed females was based on research by Cooper and colleagues (1987) that is discussed below.

The BSQ has been shown to be a valid, reliable scale. Questions were included on the BSQ based on a validity study conducted by Cooper et al. (1987) where 573 women with and without known body image disturbances were tested. Concurrent validity was demonstrated with a significant correlation ($r = 0.66$, $p < 0.05$) between the BSQ and The Body Dissatisfaction subscale of the Eating Disorder Inventory among patients with bulimia nervosa, and a correlation of $r = 0.61$ with the Eating Attitudes Test among occupational therapy students. The BSQ was also administered to 38 bulimic patients and 535 other women. All items included on the BSQ correlated at 0.60 or above for both the bulimic patients and the other women.

A second assessment of validity included two groups of women taken from community samples. The first group was made up of women who met self-report diagnostic criteria for bulimia nervosa, which included the current occurrence of bulimic episodes, compensatory vomiting or laxative use, and fear of becoming fat. The women in this group were classified as "probable cases" of bulimia nervosa. A second group, classified as "definite non-cases," did not binge, vomit, or use purgatives. These two groups differed significantly on the BSQ with the average score for "probable cases" equaling 129.3 ($SD = 17.0$) and the mean score for the "definite non-cases" equaling 71.9 ($SD = 23.6$). Lastly, in a final test for validity, two groups were taken from a non-eating-disorder patient sample. One group consisted of women concerned about weight who rated thinness as moderately to extremely important and were currently dieting. The second group consisted of women unconcerned about weight and shape and who rated thinness as not at all or slightly important and were not currently dieting.

The average BSQ score of the sample body-image-disturbed group was 109.0 with a standard deviation of 21.2. The average score for the sample unconcerned group was 55.9 with a standard deviation of 14.4. These scores are significantly different ($t = 19.6$, $p < .001$; Cooper et al., 1987). Based on these findings, participants in the present study with BSQ scores of 87.7 (109.00 – 21.20) or higher will be classified as body-image disturbed.

Procedures

All students met with the researcher three separate time periods. In the first meeting, the nature of the study was explained to potential participants, and interested volunteers gave informed consent and completed the BSQ and the BCS. In the second meeting, each participant underwent lower body and upper body strength tests to determine their pre-training strength for later determination of a training effect. The weight-training group completed the strength tests during their regularly scheduled class sessions prior to any strength training. Members of the control group scheduled a strength testing appointment. The one-repetition maximum leg press test was used to assess lower body strength. This test began with five repetitions at a lightweight. The weight was then increased for two repetitions. From this point, the weight was increased after each successfully completed repetition until the participant was no longer able to fully extend her legs (Baumgartner & Jackson, 1975). The one-repetition maximum lateral pull down test was used to assess upper body strength. Following the same procedures described above, the test began with a low weight that was gradually increased until the participant could no longer execute the movement correctly. All bars and weights were marked to ensure that the same equipment was used for both the pre- and post-tests. Strength data was collected in both absolute and relative strength scores.

Following their thirteen weeks of weight training, participants met for a third time during regularly scheduled class sessions and were tested to determine post-training one-repetition maximum for both lower and upper body strength. Members of the control group were assessed during a scheduled strength testing appointment. Prior to the strength tests, all participants completed both the BSQ and the BCS.

The experimental group completed two 50-minute resistance-training sessions with weight machines and free weights two times a week for 13 weeks as part of the university resistance-training course. Graduate-level teaching assistants knowledgeable in exercise physiology and resistance training provided instruction on strength training principles and on the proper use of the weight machines and free weights. Participants worked independently but were instructed to work all major muscles of the body, including gastrocnemius, hamstrings, quadriceps, gluteals, abdominals, pectorals, shoulders, back, bicep, and triceps using both free weights and strength training machines. The participants received instructions on how many repetitions to perform for each exercise and when to advance to a higher weight. An 80% attendance rate was required of the weight-training participants for inclusion in the study.

Statistical Analysis

A quasi-experimental design was used to examine the effect of weight training on body satisfaction among body-image-disturbed women. Several analyses were conducted to examine the nature of the data and to test the efficacy of the weight-training program on strength and body satisfaction. First, descriptive statistics and reliability coefficients were calculated. Second, two one-way ANCOVAs were conducted to examine post-training differences in absolute body strength between the two groups while controlling for pre-training strength differences. Third, a one-way ANCOVA was conducted to examine post-training differences in body satisfaction while controlling for potential pre-training group differences. ANCOVA is a statistical

technique used to control for pre-existing group differences, and the pre-treatment variables correlating highly with the dependent variable are typically the main criterion for a covariate (Keppel, 1982). Thus, correlations among the variables collected before and after the treatment were examined to confirm the appropriateness for the use of this analysis. Lastly, the body image disturbance classification was re-examined at the end of the study. As with the initial classification, a minimum value of 87.7 was used to identify body-image-disturbed participants. An alpha level of .10 was used for all statistical tests because the risk associated with a Type I error was deemed minimal relative to the risks of committing a Type II error.

RESULTS

Internal Consistency and Correlations

The internal consistency scores associated with body satisfaction was high with Cronbach alpha coefficients of .91 and .94 on the pre- and post-training satisfaction scores, respectively. Similar internal consistency scores were found for body image disturbance with alpha coefficients of .86 and .94 on the pre- and post-training body image disturbance scores, respectively. Correlations among the corresponding covariate and dependent variable ranged from .78 to .90, thus confirming the appropriateness of using ANCOVAs.

Manipulation Check

A manipulation check was conducted to determine the efficacy of the weight-training program in improving strength. Using pre-test lateral pull down scores as a covariate and treatment groups as the independent variable, an ANCOVA revealed statistically significant differences in upper-body strength between the weight training and the control groups following the weight training treatment (see Table 1). The weight-training group experienced greater strength gains than the control group. Pre and post scores can be seen in Table 2. A second ANCOVA was conducted with pre-test leg press scores as the covariate and also demonstrated a statistically significant increase in lower body strength for the weight-training group when compared to the control group (see Tables 1 and 2). Together, these results demonstrate that the thirteen-week weight-training program was effective in improving upper and lower body strength.

Table 1
Analysis of Covariance for Strength and Body Image

Source	<i>df</i>	<i>F</i>	<i>Eta</i> ²	<i>p</i>
Upper body strength				
Pretest lateral pull down	1	73.07	.73	<.001
Weight training effect	1	35.45	.57	<.001
Lower body strength				
Pretest leg press	1	82.55	.75	<.001
Weight training effect	1	16.36	.38	<.001
Body satisfaction				
Pretest body satisfaction	1	126.14	.82	<.001
Weight training effect	1	5.67	.17	.025
Body image disturbance				
Pretest body image	1	102.96	.79	<.001
Weight training effect	1	6.22	.19	.019

Table 2
Pre and Post Means and Standard Deviations (in Parenthesis) for
Strength and Body Image Scores

	Pre scores		Post scores	
	WT group	Control group	WT group	Control group
Strength				
Upper body	79.33 (19.45)	70.67 (5.94)	89.33 (14.86)	69.33 (7.04)
Lower body	287.67 (68.58)	346.00 (55.27)	350.67 (73.70)	342.00 (57.60)
Body image				
Satisfaction	3.19 (.56)	3.10 (.51)	3.40 (.63)	3.10 (.49)
Disturbance	107.33 (17.79)	126.53 (26.18)	95.33 (22.65)	126.00 (27.17)

Note. WT = weight training.

Efficacy of Treatment

With pre-training body satisfaction scores as the covariate, an ANCOVA was conducted to determine the effects of weight training on body satisfaction. Results revealed a statistically significant increase in body satisfaction scores among the weight training participants when compared to those in the control groups following the thirteen-week weight training treatment (see Tables 1 and 2). Reexamination of the classification of individuals following the study, revealed that seven of the 15 (~41%) weight training, and only 1 of the 15 (~7%) non-weight training participants experienced reductions in their negative body image scores such that they were no longer classified as body-image-disturbed. An ANCOVA using pre-training body image disturbance as the covariate, revealed a statistically significant difference between the weight training group and the control group at the close of the thirteen-week weight-training program (see Table 1). Specifically, the weight-training group experienced a greater reduction in body image disturbance than the control group (see Table 2).

DISCUSSION

The primary purpose of this study was to examine the association between weight training and body satisfaction levels of college-aged women with body image disturbances. It was hypothesized that body-image-disturbed college females in a weight-training program were more likely to improve body satisfaction than body-image-disturbed college females not participating in a weight-training program. Overall, support for this hypothesis was found.

The results of this study suggest that weight training was related to increases in body satisfaction levels of body-image-disturbed college women. Specifically, those who participated in the thirteen-week weight training class reported a greater increase in body satisfaction than those who did not participate in the weight-training program. Additionally, results revealed that weight training was negatively associated with body image disturbance. Females who participated in a weight training class scored lower on body image disturbance than those in the control group. In fact, at the end of the thirteen weeks, more than one-third of those in the weight-training group were no longer classified as body-image disturbed.

Results regarding weight training found in the current study are consistent with and extends the results of previous research conducted with males (James, 1982; Tucker, 1982a, 1982b, 1987) and females (Caruso & Gill, 1992; Cullinen & Caldwell, 1998; Holloway, Beutter, & Duda, 1988; Tucker & Maxwell, 1991). Researchers using participants from weight training and general education classes found that males participating in weight training twice a week for sixteen weeks reported a more positive body image than those who did not. Similarly, studies have demonstrated that women who participated in weight training, some for as little as ten weeks, showed significantly greater gains in body image scores than women who did not participate. The current study extends the knowledge base of the relationship between weight training and body satisfaction by focusing on body-image-disturbed females. Collectively, the results provide support for the contention that weight training may impact the body image of men and women in general and are related to increases in body satisfaction among women who are body-image disturbed.

In contemplating why successful weight training may contribute to enhanced body satisfaction, it is reasonable to consider the effects of weight training on self-perceptions. Sonstroem (1997) contends that physical activity participation leads to increased physical ability resulting in enhanced psychological benefits. Fox and Corbin (1989) suggest that physical activity impacts our physical self-perceptions, such as perceived physical strength and conditioning, which in turn impacts our physical self-worth and global self-esteem. Although the current study did not assess perceived strength, individuals participating in the weight-training program did get stronger. The results of the present study parallel previous research findings revealing that females who engaged in weight training demonstrated significant increases in total muscular strength, significant decreases in total skinfold thickness, and enhanced body satisfaction (Caruso & Gill 1992; Tucker & Mortell, 1991). Based on the research findings of this study and previous studies, in combination with self-esteem theories, it is plausible that the changes in body satisfaction among body-image-disturbed women involved in weight training may be a result of greater actual and perceived strength with accompanying increases in physical self-perceptions and physical self-worth.

Regardless of the mechanisms associated with changes in body satisfaction, the results of the present study provide support for the contention that at least two days of weight training over the course of thirteen weeks is associated with increased strength, greater body satisfaction, and reduced body image disturbance. These findings provide initial evidence that weight training may serve as a method of treatment for body image disturbance and may work effectively in conjunction with currently used cognitive therapies. In addition, weight training may be a particularly effective treatment due to the immediate feedback of weight training as the training load is progressively increased to match the body's improved strength (Tucker, 1982a, 1982b, 1987). This, along with the observable positive changes in the body's physique, may provide individuals with positive and concrete information regarding changes in their appearance.

What makes weight training especially appealing as a treatment is its contribution to overall health, well-being, and fitness (American College of Sports Medicine, 1995) and its easy accessibility. Given the accessibility of weight training via university recreation centers, health clubs and apartment complexes; the increased availability of affordable fitness equipment for the home; and the adaptability of common home materials for use in weight training, weight training can be a cost effective treatment for body-image-disturbed college females with the added bonus of enhanced strength, fitness, and health.

As we continue to consider the effectiveness of weight training as a potential therapy for body-image-disturbed women, it is important to consider the limitations of research and implications for future research in this area. Despite the use of the statistical procedures to control for pre-existing differences in body satisfaction and body image disturbance, participants in

the current study, as in several other studies, were recruited from intact weight training classes, resulting in non-random sampling. In essence, participants in the weight training group self-selected involvement in weight training. It is plausible that some individuals may be more motivated to engage in proper weight training techniques and thus are more likely to reap the benefits of strength training with concurrent improvement in body image and satisfaction than those not motivated to engage in weight training. A person who voluntarily begins a weight-training class may also be more likely to alter eating and activity habits. For example, the participants in this study were body-image-disturbed females who exhibited a readiness for strength training and therefore may have been more likely to experience increases in body satisfaction and reductions in body image disturbance than those not ready for routine weight training sessions. Thus, the results of the current study indicate that weight training is effective for those demonstrating a readiness to weight train. To understand more fully the effects of weight training on body satisfaction and image disturbance, future research should consider the role of self-selection in the effectiveness of weight training on body image-related factors.

Two other limitations include the possible implications of the differences between the weight training and control groups in regard to age and ethnic composition. The mean age among the weight training group was slightly higher and associated with greater variability than the control group. Research has demonstrated that the incidence of body image concerns often peak during college-age years (Davis et al., 1990) leading some to assume that older women may have less severe body concerns. However, research has consistently shown that physical appearance is a salient feature of self-esteem throughout the lifespan (Loland, 2000) and that body satisfaction does not differ across age groups (Altabe & Thompson, 1990; Davis & Cowles, 1991; Pilner, Chaiken, & Flett, 1990). Moreover, age has been shown to be unrelated to changes in general body dissatisfaction, as was measured in the current study (Dionne, Davis, Fox, & Gurevich, 1995).

The weight-training group was more racially diverse than the predominantly Caucasian control group. Studies have indicated that Caucasian women have a higher incidence of body image concerns than other ethnic groups. Women of color are not as likely to identify with the "American Ideal" and therefore may have a healthier opinion of their own bodies (Parker & Nichter, 1995). Since the weight training treatment was applied to a more racially diverse group, these results may differ from what may be seen with other body-image-disturbed females who identify more with societal pressures to attain a certain appearance. Interestingly, one can only speculate as to whether the improvement in body image occurred because the weight training group had a healthier opinion of their bodies, or whether body-image-disturbed Caucasian females would tend to be even more satisfied with weight training effects because of the societal norm associated with the "American Ideal." Future research should consider the role of race and culture when considering the effects of exercise on body image.

Furthermore, future research needs to be conducted to determine the optimal length of the program, number of days per week, and length of each training session of a weight-training program for improving body satisfaction and body image disturbance. Studies have demonstrated improvements in body image as a result of programs lasting for 10–16 weeks, 2–3 days per week, and incorporating 50–90 minute workout sessions (Caruso & Gill, 1992; Folkins, 1976; Holloway et al., 1988; James, 1982; Tucker, 1982a, 1982b, 1983, 1987; Tucker & Maxwell, 1991; Tucker & Mortell, 1993). However, it is not evident what training period optimizes body image enhancement. Thus, it may be beneficial to conduct research in which the length of the weight training program, the number of days spent lifting per week, and duration of each weight training session are manipulated.

A need also exists for studies examining the lasting effects of weight training on body image satisfaction. The present research demonstrated that weight training was positively associated

with increased body satisfaction of body-image-disturbed college females immediately following the weight-training program. Longitudinal studies that focus on changes in body image disturbance and body satisfaction once training ceases may provide information concerning the long-term effects of weight-training among individuals with body image-disturbance.

Finally, treatment for body image disturbances will be advanced by determining the effectiveness of using cognitive therapy in conjunction with a weight-training program, particularly for college age women with body image disturbances. Unfortunately, no research exists studying the combined effects of cognitive therapy and weight training on body image disturbance and satisfaction.

In conclusion, the results of the present study revealed that body-image-disturbed college women participating in a weight training class for thirteen weeks demonstrated increased body satisfaction, upper and lower body strength, and reduced body image disturbance. At the end of the study, more than one-third of the weight-training females were no longer classified as "body-image-disturbed." Information from the present study provides a foundation for valuable insights for practitioners working with college women with body image disturbance. Further research in the area of weight training as a treatment for body image disturbance, especially when combined with cognitive treatments, is likely to provide important information for enhancing the health and well being of individuals with body image disturbances.

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